

Demonstration of Sympathetic Orchestra: An Interactive Conducting Education System for Responsive, Tacit Skill Development

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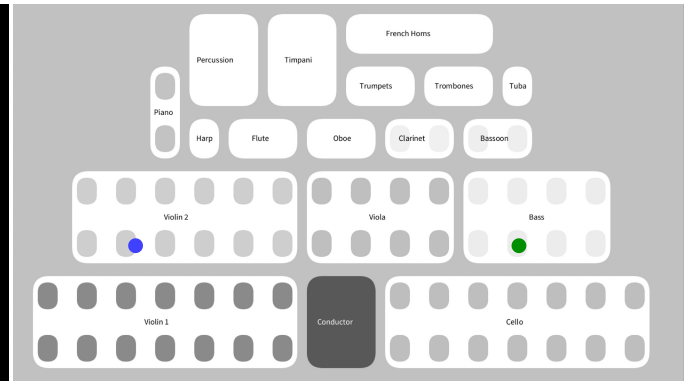


Figure 1: SYMPATHETIC ORCHESTRA is a dynamic virtual symphony orchestra, featuring an interface designed to mirror the layout of a real symphony orchestra. Users can use their hands and faces to direct and can receive responsive feedback.

ABSTRACT

Students learning musical conducting often practice along to static recordings, which do not provide real-time feedback similar to that of a live orchestra during rehearsals. Novice conductors need better solutions for practicing with feedback that mimics the experience of conducting a live orchestra. We can leverage emergent multimodal and spatial interaction technologies to support a “virtual orchestra” practice experience that allows students to develop tacit, live-practice knowledge. Through formative interviews with conducting experts and students, we designed and developed a dynamic, multimodal interaction system that targets key goals held by students developing their orchestral conducting skills, and that traditional practicing methods lack support for. Sympathetic Orchestra is an interactive virtual orchestra system that uses Google AI edge-powered Computer Vision hand and face tracking on webcam data

to responsively interact with dynamic audio music playback and develop tacit practicing experiences.

CCS CONCEPTS

• Human-centered computing → Interaction paradigms.

KEYWORDS

Interaction Design, Conducting and Music Education

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1 INTRODUCTION

Conducting a symphony orchestra is a professional job with a well-established educational system and a long history. In conducting education, where students are training to conduct a symphony orchestra, there are significant gaps between the student learning experience and actual live performance experiences. Although the methods commonly used in the educational process (playing static

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recordings or conducting to piano players) can roughly simulate the experience of conducting a symphony orchestra to a certain extent, the (visual, audio, and spatial) limitations of these methods impede students' ability to develop the tacit knowledge, and confidence necessary to conduct a real symphony orchestra. These methods also impede the development of higher-level creative expression and interpretive exploration, and instead tend to create a static, one-dimensional practicing experience that is inaccurate to the responsive, modular, spatially and temporally dynamic experience of conducting an orchestra.

Emerging interaction audio computing technologies, as well as rapid advancements in computer vision offer new possibilities for supporting students learning to conduct. Our system offers an intuitive, accessible solution to supplementing this gap.

In this demonstration, we present SYMPATHETIC ORCHESTRA, an educational creativity support system designed to assist in conducting education and practice. Users can use their hands and faces to direct a dynamic virtual symphony orchestra, featuring an interface designed to mirror the layout of a real symphony orchestra. They can receive the responsive feedback necessary to develop tacit knowledge, with features that support understanding, practice, error reminder, as well as the opportunity to develop higher-level creative expression and style in an interactive musical playback experience that dynamically responds to their interpretive choices as a conductor.

2 BACKGROUND & MOTIVATION

Our work builds on previous research[11][7][12][10][9][17][2][13][16], educational theories[1][8][20][5][14][3][6], and design frameworks[21][15][19]. The Inspect, Embody, Invent Design Framework for Music Learning[21] in particular motivated the design of a system that would first support a conducting student in understanding a piece, then seamlessly continue to support them as they start to embody, invent, and creatively interpret. Prior studies have explored how virtual orchestras that simulate the timbre and feedback of real symphony orchestras can effectively help conductors practice[11][7][12][10][9]. Sarasua et al. used computer vision supported gesture recognition for controlling the mixed audio playback of different orchestral sections[17]. Ivanova et al. explored how visual feedback can help students better learn a piece during the practicing process [9]. BeaudoinLafon et al apply Goodwin's concept of professional vision to color grading to explore how a system can be designed to support individuals developing professional vision[8][4]; the design of SYMPATHETIC ORCHESTRA lends from experts in conducting education to support the development of "professional hearing".

On the surface, achieving these functions may seem as simple as mixing in a Digital Audio Workstation(DAW), but it can be a challenge to design educational tools and processes that effectively build tacit skills useful in real-world practices. We designed this interactive system using educational theories from Professional Vision[8] and the Inspect, Embody, Invent design framework[21], making the system an effective tool for assisting in conducting education and practice. We then developed the system using p5.js for an accessible web-based interface, Google AI Edge for real-time hand and face tracking and gesture recognition, and Web Audio APIs for rich, dynamically responsive audio playback and

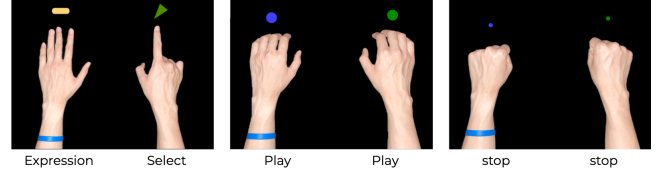


Figure 2: Gestures and corresponding cursors.

Function (Action)	Description
EXPRESSION(<i>palm spread</i>)	Changes the modulation of the music, such as dynamics and expressiveness.
SELECT(<i>finger point</i>)	Selects a specific part.
PLAY(<i>relax naturally</i>)	Continues playing.
STOP(<i>fist</i>)	Stops playing.
TEMPO(<i>conducting gestures</i>)	Controls the tempo according to the score and corresponding beat.
CUEING(<i>cursor appears one beat in advance in the corresponding part</i>)	Cues the part that is about to start playing.

Table 1: Available functions in the prototype and their corresponding actions and explanations. The actions control the audio, providing real-time sound feedback during the conducting practice process.

mixing. We plan to evaluate the system's effectiveness through micro-analysis[18] of the students' multimodal utterances as they explain and use the system, assessing the cognitive ergonomics of the system in the students' intuitive perception-action loops, and examining how the sound output serves as feedback shaping their skill development.

3 LEARNING AND PRACTICING WITH SYMPATHETIC ORCHESTRA

SYMPATHETIC ORCHESTRA uses a webcam and computer vision to recognize gestures and process audio to simulate the realistic sound feedback of a symphony orchestra. The interface provides responsive visual feedback, a feature that existing conducting education and practice methods lack.

To help understand the music, our prototype also provides background information about the music, visual feedback during the practice process, and recording and playback functions.

4 USAGE SCENARIO

To motivate our prototype, consider the following scenario:

Herbert is an undergraduate student studying symphony orchestra conducting. He wants to perform Rachmaninoff Piano Concerto No.2 with his pianist friend Alexis in an upcoming concerto competition. Before rehearsing with the actual orchestra, he hopes to hear the sound of the orchestra collaborating with the piano to help

him think about further musical processing. He hopes this practice process will help him prepare to conduct a real orchestra.

Getting Started. Herbert and Alexis come to the practice room. Herbert connects the computer to the stereo monitor speakers in the practice room and selects Rachmaninoff Piano Concerto No.2. The screen displays the background of the work composed by Rachmaninoff, and Herbert learns that the piece was dedicated to the psychiatrist who saved the composer.

Practicing. Herbert begins conducting the Sympathetic Orchestra while Alexis plays the piano. Initially, the visual effects on the interface indicate the parts Herbert forgot to cue. As Herbert practices, he develops tacit knowledge of the piece and becomes more familiar and confident with its movements. He practices the body movements in his hands, arms, and face that he learned in class to cue and direct each section of the orchestra in time with the music.

Iterating and Sharing. As Herbert becomes more and more familiar with this piece, he begins to experiment with his creative interpretation. Herbert and Alexis have several different creative ideas for handling the piece and want to compare them, as they consider what adjustments might guide the orchestra to collaborate with the piano. The Sympathetic Orchestra interactively responds to Herbert's movements as they try different tempos and modulations for different orchestral sections. At times they choose to highlight the piano, and at other times, the clarinet; they listen to the mixed harmony as they experiment with signaling different timbres for the wind and string instruments, attempting to evoke a variety of emotions throughout the piece. Herbert uses the recording and playback functions to compare five different handling methods across more than 20 sections, eventually finding an approach that he believes best expresses his musical understanding while achieving a delicate balance with the piano, and preparing him for rehearsal with a live symphony orchestra.

5 DEMONSTRATION SETUP

Our demonstration setup includes a laptop running the prototype interface, a webcam, and a set of stereo speakers, configured with several usage examples and tutorial sheets.

6 CONCLUSION

We demonstrate an interactive system designed to assist in professional conducting education and practice. Built on p5.js, Google AI Edge, and Web Audio APIs and informed by theoretical frameworks and formative interviews with conducting experts and students, our prototype offers a new solution for conducting education. It uses computer simulation to create a dynamic, responsive multi-model spatial practice experience to better develop skills to prepare students for conducting a live symphony orchestra. In this demonstration, users can explore the art of conducting on their own computer.

REFERENCES

- [1] Dor Abrahamson and Rotem Abdu. 2021. Towards an ecological-dynamics design framework for embodied-interaction conceptual learning: The case of dynamic mathematics environments. *Educational Technology Research and Development* 69 (2021), 1889–1923.
- [2] Juan Pablo Martinez Avila, Chris Greenhalgh, Adrian Hazzard, Steve Benford, and Alan Chamberlain. 2019. Encumbered Interaction: a Study of Musicians Preparing to Perform. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3290605.3300706>
- [3] Jeanne Bamberger and Andrea DiSessa. 2003. Music as embodied mathematics: A study of a mutually informing affinity. *International Journal of Computers for Mathematical Learning* 8 (2003), 123–160.
- [4] Matthew T Beaudouin-Lafon, Jane L E, and Haijun Xia. 2023. Color Field: Developing Professional Vision by Visualizing the Effects of Color Filters. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) (UIST '23). Association for Computing Machinery, New York, NY, USA, Article 101, 16 pages. <https://doi.org/10.1145/3586183.3606828>
- [5] Daniel Black. 2014. Where bodies end and artefacts begin: Tools, machines and interfaces. *Body & Society* 20, 1 (2014), 31–60.
- [6] Paul Cobb, Jere Confrey, Andrea DiSessa, Richard Lehrer, and Leona Schauble. 2003. Design experiments in educational research. *Educational researcher* 32, 1 (2003), 9–13.
- [7] Joyce Horn Fonteles and Maria Andréia Formico Rodrigues. 2021. User experience in a kinect-based conducting system for visualization of musical structure. *Entertainment Computing* 37 (2021), 100399. <https://doi.org/10.1016/j.entcom.2020.100399>
- [8] Charles Goodwin. 2015. Professional vision. In *Aufmerksamkeit: Geschichte-theorie-empirie*. Springer, 387–425.
- [9] Ekaterina Ivanova, Lulu Wang, Yihe Fu, and Jeffrey Gadzala. 2014. MAES:TRO: a practice system to track, record, and observe for novice orchestral conductors. In *CHI '14 Extended Abstracts on Human Factors in Computing Systems* (Toronto, Ontario, Canada) (CHI EA '14). Association for Computing Machinery, New York, NY, USA, 203–208. <https://doi.org/10.1145/2559206.2580929>
- [10] Eric Lee, Henning Kiel, Saskia Dedenbach, Ingo Grüll, Thorsten Karrer, Marius Wolf, and Jan Borchers. 2006. iSymphony: an adaptive interactive orchestral conducting system for digital audio and video streams. In *CHI '06 Extended Abstracts on Human Factors in Computing Systems* (Montréal, Québec, Canada) (CHI EA '06). Association for Computing Machinery, New York, NY, USA, 259–262. <https://doi.org/10.1145/1125451.1125507>
- [11] Eric Lee, Teresa Marrin Nakra, and Jan Borchers. 2004. You're the conductor: a realistic interactive conducting system for children. In *Proceedings of the 2004 conference on New interfaces for musical expression*. 68–73.
- [12] Eric Lee, Marius Wolf, and Jan Borchers. 2005. Improving orchestral conducting systems in public spaces: examining the temporal characteristics and conceptual models of conducting gestures. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Portland, Oregon, USA) (CHI '05). Association for Computing Machinery, New York, NY, USA, 731–740. <https://doi.org/10.1145/1054972.1055073>
- [13] Kyungho Lee, Donna J. Cox, Guy E. Garnett, and Michael J. Junokas. 2015. Express it! An Interactive System for Visualizing Expressiveness of Conductor's Gestures. In *Proceedings of the 2015 ACM SIGCHI Conference on Creativity and Cognition* (Glasgow, United Kingdom) (C&C '15). Association for Computing Machinery, New York, NY, USA, 141–150. <https://doi.org/10.1145/2757226.2757243>
- [14] Luciano Meira. 1998. Making sense of instructional devices: The emergence of transparency in mathematical activity. *Journal for Research in Mathematics Education* 29, 2 (1998), 121–142.
- [15] Dan R Olsen Jr. 2007. Evaluating user interface systems research. In *Proceedings of the 20th annual ACM symposium on User interface software and technology*. 251–258.
- [16] Elizabeth B.-N. Sanders and Pieter Jan Stappers. 2008. Co-creation and the new landscapes of design. *CoDesign* 4, 1 (2008), 5–18. <https://doi.org/10.1080/15710880701875068> arXiv:<https://doi.org/10.1080/15710880701875068>
- [17] Alvaro Sarasua, Baptiste Caramiaux, and Atsuo Tanaka. 2016. Machine Learning of Personal Gesture Variation in Music Conducting. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '16). Association for Computing Machinery, New York, NY, USA, 3428–3432. <https://doi.org/10.1145/2858036.2858328>
- [18] Robert S Siegler and Kevin Crowley. 1991. The microgenetic method: A direct means for studying cognitive development. *American psychologist* 46, 6 (1991), 606.
- [19] Victor Udoewa. 2022. An introduction to radical participatory design: decolonising participatory design processes. *Design Science* 8 (2022), e31. <https://doi.org/10.1017/dsj.2022.24>
- [20] Pierre Verillon and Pierre Rabardel. 1995. Cognition and artifacts: A contribution to the study of thought in relation to instrumented activity. *European journal of psychology of education* (1995), 77–101.
- [21] Xiao Xiao and Hiroshi Ishii. 2016. Inspect, Embody, Invent: A Design Framework for Music Learning and Beyond. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '16). Association for Computing Machinery, New York, NY, USA, 5397–5408. <https://doi.org/10.1145/2858036.2858577>